Building Prediction Model

Python course - Lesson 8

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Previous Lesson Topics

- Introduction to Seaborn package
- Select the correct graph type
- Graph types
- Change style od the graph

Today's Topics

- Introduction to Scipy, statsmodels and sklearn packages
- What is a model
- Steps when building a model

Introduction to Machine Learning Packages

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Packages For Machine Learning

SciPy

statsmodels

scikit learn

What is SciPy

- SciPy is an open-source Python library used to solve scientific and mathematical problems
- SciPy contains fully-featured versions functions that can help resolve linear algebra, Fourier transforms, etc
- SciPy builds on NumPy, therefore if you import SciPy, there is no need to import NumPy (but you need to install both NumPy and SciPy)

StatsModels

- StatsModels is a Python package that allows users to explore data, estimate statistical models, and perform statistical tests
- StatsModels is part of the Python scientific stack that is oriented towards data analysis, data science and statistics
- StatsModels contains extensive list of descriptive statistics, statistical tests, plotting functions, and result statistics

Scikit-learn (Sklearn)

- Scikit-learn is a Python library containing features as: classification, regression and clustering algorithms including support vector machines, random forests, gradient boosting, k-means and DBSCAN

- Scikit-learn integrates well with many other Python libraries, such as matplotlib, NumPy, pandas DataFrame, SciPy, and many more

Introduction to Machine Learning

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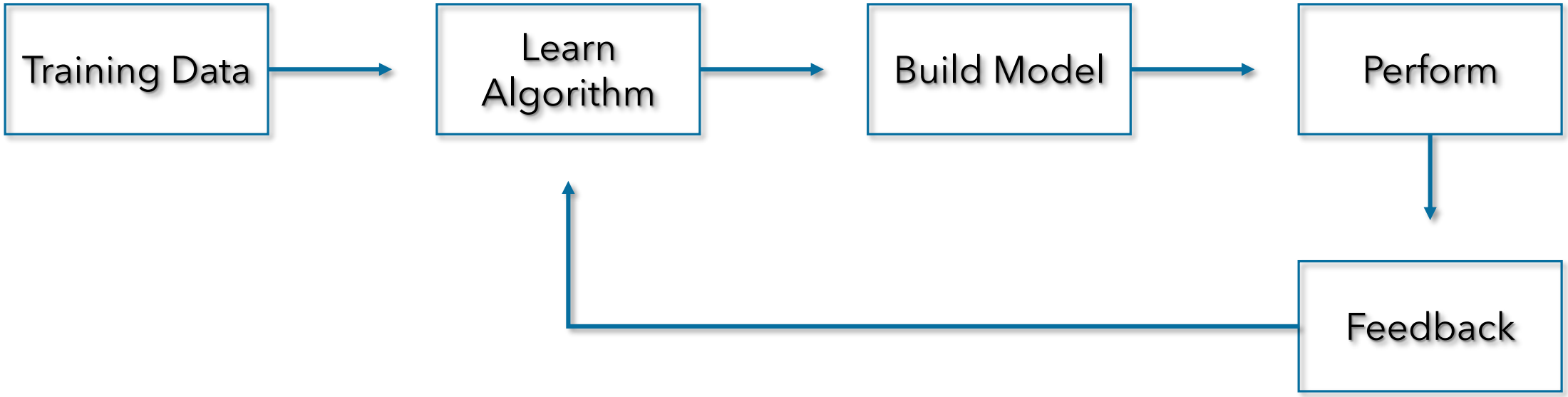
What is Machine Learning?

- Machine learning is a type of artificial intelligence that allows software applications to learn from the data and become more accurate in predicting outcomes without human intervention
- the machine needs to be trained on some data and based on that, it will detect a pattern to create a model

Machine Learning Process



3 Types of Machine Learning

Supervised Learning

Making predictions using the data

The algorithms include:

- Linear Regression
- Logistic Regression
- Decision tree
- Random forest

Unsupervised Learning

Grouping objects based on the information in the data

The algorithms include:

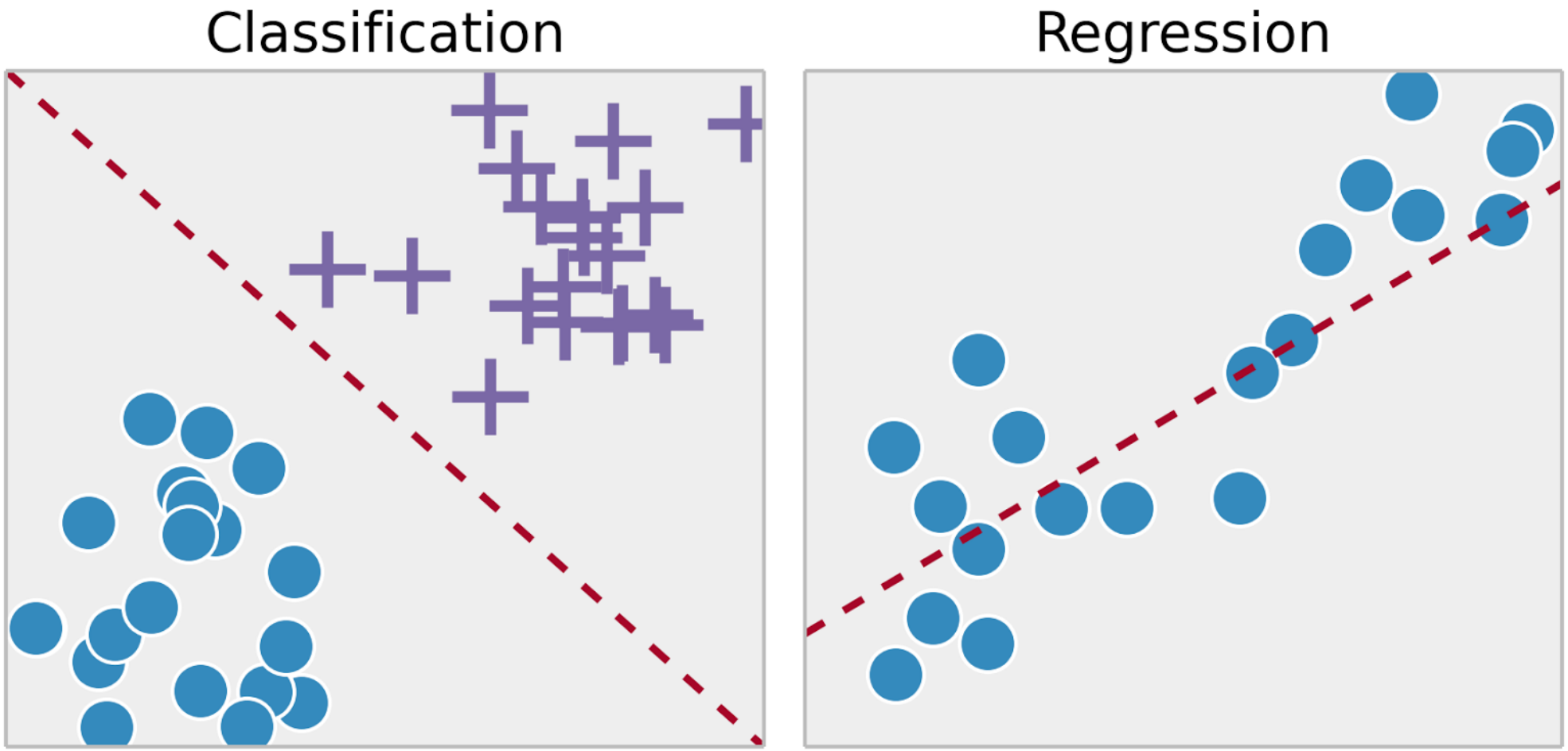
- K-means clustering
- Hierarchical clustering

Reinforcement learning

An RL agent learns from the consequences of its actions

It selects its actions on basis of its past experiences (exploitation) and also by new choices (exploration)

Supervised Algorithms



Classification

- Samples belong to two or more classes
- We want to learn from already labeled data on how to predict the class of unlabeled data
- Classification is when you have limited number of categories.

  You get n samples, and trying to label them with the correct category

Regression

- When the output consists of one or more continuous variables, then the task is called regression

- regression analysis is a set of statistical processes for estimating the relationship between one dependent variable (usually denoted by Y) and one or more independent variables

Introduction to Models

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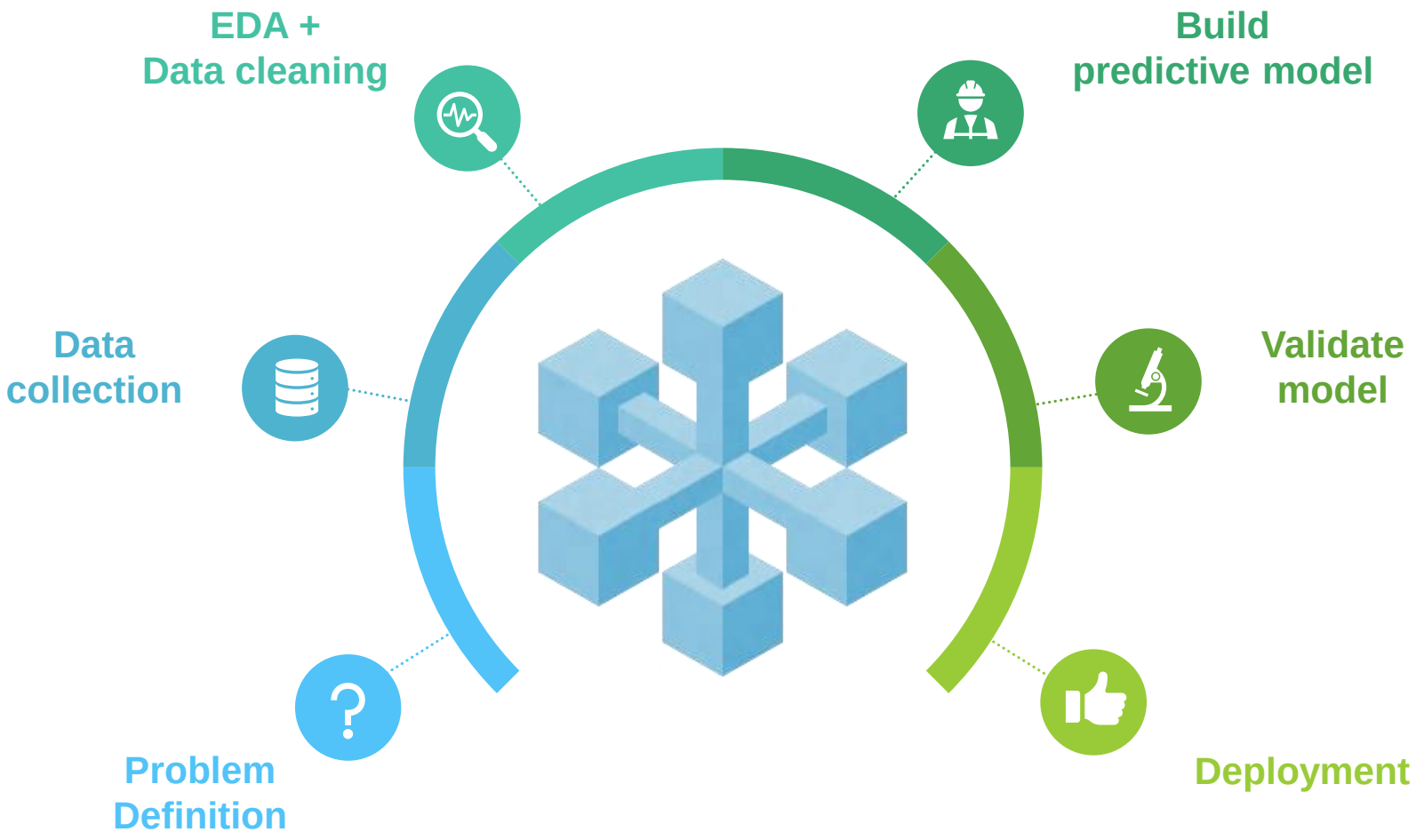
What is a Model?

- A machine learning model is a file that has been trained to recognize certain types of patterns
- You train a model over a set of data, providing it an algorithm that it can use to reason over and learn from those data
- Once you have trained the model, you can use it to reason over data that it hasn't seen before, and make predictions about those data

The Process of Building a Model



Predictions With linear Regression Model

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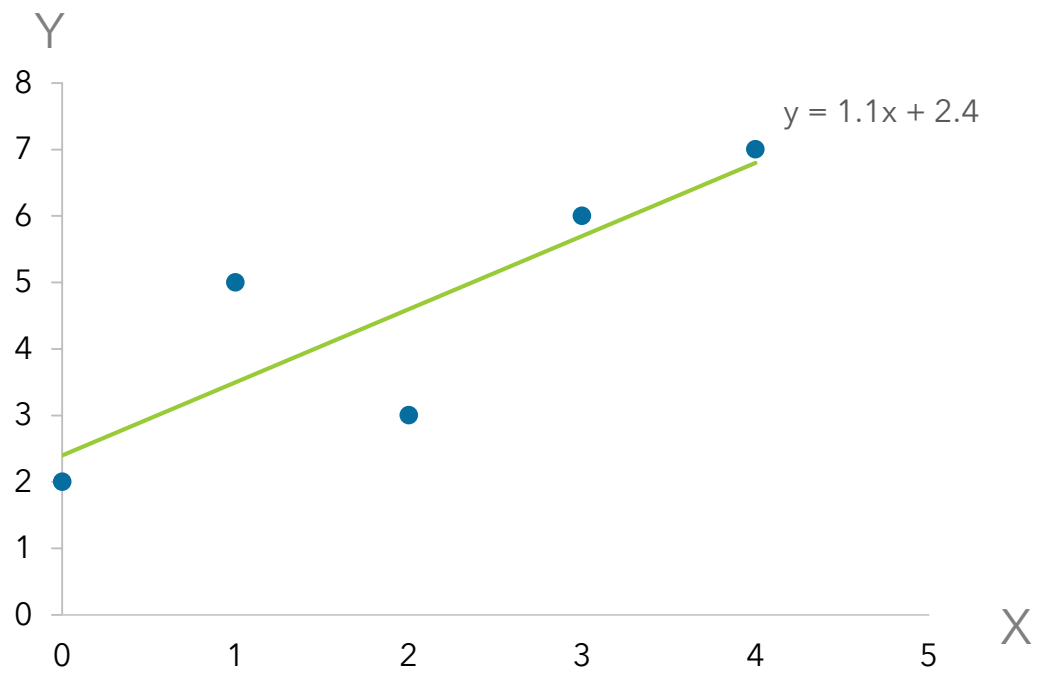
Linear Regression Formula

- One Feature:



y intercept
(level of y when x is 0)

$$f(x) = \hat{y} = b_0 + b_1 * x$$

y = Predicted values of y

b = coefficient = slop of the regression line

Linear Regression Formula

- Multiple Features

y intercept
(level of y when x is 0)

$$f(x) = \hat{y} = b_0 + b_1 * x_1 + b_2 * x_2 + \cdots + b_n * x_n$$

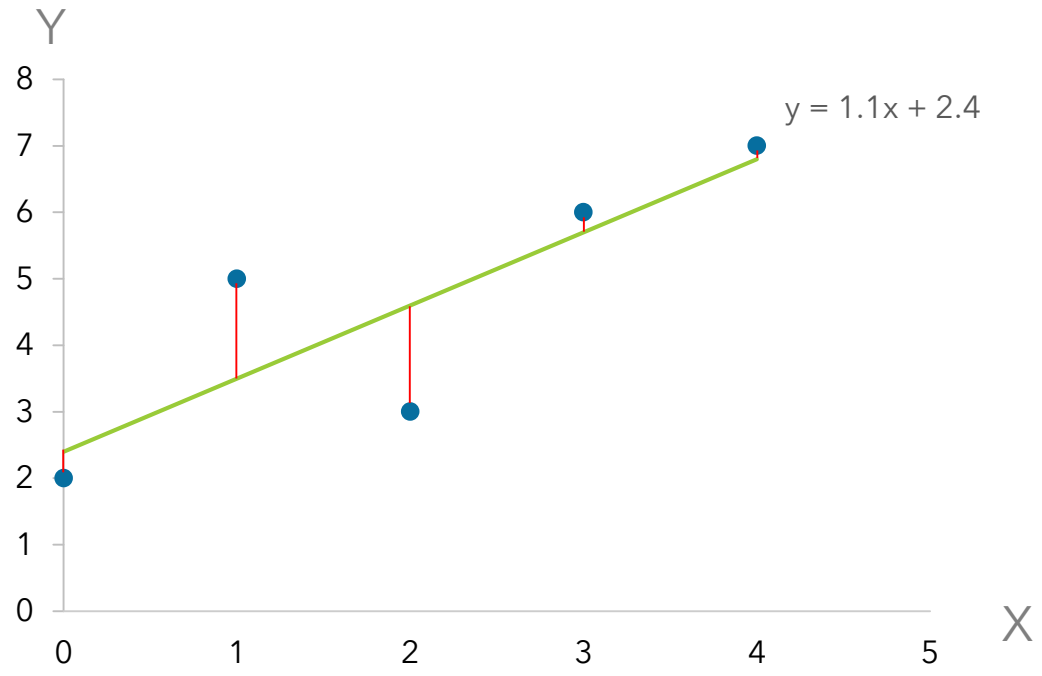
y = Predicted values of y

b = coefficients = significance of effect on the dependent variable

MSE - Mean Squared Error

- MSE of an estimator measures the average of the squares of the errors, that is, the average squared difference between the estimated values and the actual value

- The lower the value, the better



$$MSE = \frac{1}{n} \sum_{i=1}^{n} \underbrace{(y_i - \hat{y}_i)^2}_{\text{square of the difference between actual and predicted}}$$

Thank You

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